

OPERATING SYSTEMS (CS304)

File Allocation Methods

Learning Objectives

Understand file allocation techniques, storage organization, and the advantages and disadvantages of different allocation methods.

Introduction to File Systems

A file system organizes and manages data stored on secondary storage devices. File allocation methods determine how files are stored and accessed on disks.

What is File Allocation?

File allocation is the process of assigning disk blocks to files. The operating system maintains information about the location of file blocks.

Objectives of File Allocation

- Efficient disk space utilization
- Fast file access
- Easy file management
- Reduced storage fragmentation

Contiguous Allocation

In contiguous allocation, file blocks are stored in consecutive disk locations. The operating system stores the starting block and file length.

Advantages of Contiguous Allocation

- Simple implementation
- Fast sequential access
- Fast direct access

Disadvantages of Contiguous Allocation

- External fragmentation
- Difficult file expansion
- Space wastage

Linked Allocation

In linked allocation, each file consists of linked disk blocks. Every block contains a pointer to the next block.

Advantages of Linked Allocation

- No external fragmentation
- Easy file growth
- Flexible storage utilization

Disadvantages of Linked Allocation

- Slow direct access
- Pointer overhead
- Reliability issues if links are damaged

Indexed Allocation

Indexed allocation stores all block addresses in a separate index block. The index block contains pointers to all file blocks.

Advantages of Indexed Allocation

- Supports direct access
- No external fragmentation
- Efficient file growth

Disadvantages of Indexed Allocation

- Additional storage for index blocks
- More complex implementation

Comparison of Allocation Methods

Contiguous: Fast access but fragmentation issues.

Linked: Flexible but slower access.

Indexed: Efficient access with additional index overhead.

Applications

Modern operating systems use variations of indexed allocation techniques to efficiently manage large file systems.

Exam Important Questions

1. Define File Allocation.
2. Explain Contiguous Allocation.
3. Explain Linked Allocation.
4. Explain Indexed Allocation.
5. Compare File Allocation Methods.

Summary

File allocation methods determine how files are stored on disk. Contiguous, Linked, and Indexed allocation each provide unique advantages and trade-offs in terms of access speed, storage efficiency, and flexibility.